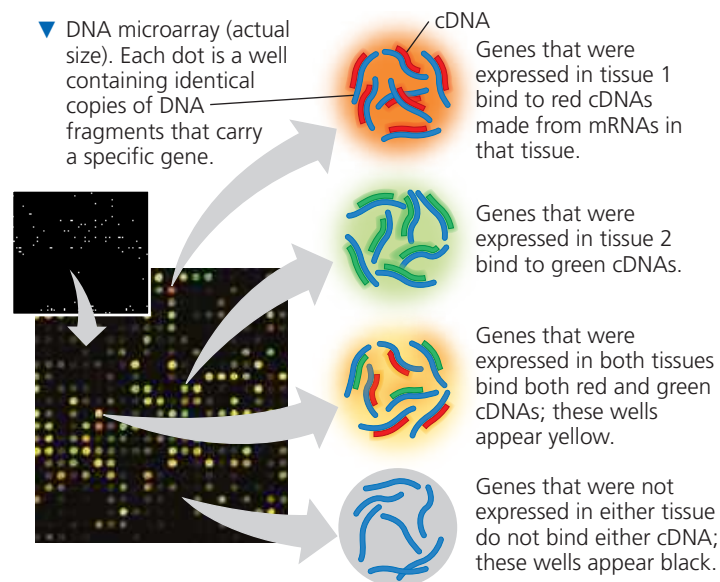


Several characteristics make RNA-seq a powerful technique. First, it is not based on hybridization with a labeled probe, so it doesn't depend on knowing genomic sequences. Second, it can measure levels of expression over a very wide range. Third, a careful analysis provides a wealth of information about expression of a particular gene, such as relative levels of alternatively spliced mRNAs. In most cases, however, expression of individual genes still needs to be confirmed by RT-PCR.

An older method of genome-wide expression studies, called **DNA microarray assays**, is less powerful than RNA-seq but is still used for some applications, such as fetal testing (see Figure 14.19). A DNA microarray consists of tiny amounts of a large number of single-stranded DNA fragments representing different genes fixed to a glass slide in a tightly spaced array, or grid, of dots. (The microarray is also called a *DNA chip* by analogy to a computer chip.) Ideally, these fragments represent all the genes of an organism. The mRNAs from the cells being studied are reverse-transcribed into cDNAs (see Figure 20.10), and a fluorescent label is added so the cDNAs can be used as probes on the microarray. Different fluorescent labels are used for different cell samples so that multiple samples can be tested in the same experiment. The resulting pattern of colored dots, shown in an actual-size microarray in **Figure 20.13**, reveals the dots to which each probe was bound and thus the genes that are expressed in the cell samples being tested.

▼ **Figure 20.13 Use of microarrays to analyze expression of many genes.** Researchers extracted mRNAs from two different human tissues and synthesized two sets of cDNAs, fluorescently labeled red (tissue 1) or green (tissue 2). Labeled cDNAs were hybridized with a microarray containing 5,760 human genes (about 25% of human genes), part of which is shown in the enlargement. Red indicates that the gene in that well was expressed in tissue 1, green in tissue 2, yellow in both, and black in neither. The fluorescence intensity at each spot indicates the relative expression of the gene.



Scientists can now measure the expression of thousands of genes at one time. DNA technology makes such studies possible; with automation, they are easily performed on a large scale. By uncovering gene interactions and providing clues to gene function, DNA microarray assays and RNA-seq may contribute to a better understanding of diseases and suggest new diagnostic techniques or therapies. For instance, comparing patterns of gene expression in breast cancer tumors and noncancerous breast tissue has already resulted in more informed and effective treatment protocols (see Figure 18.27). Ultimately, information from these methods should provide a grander view of how ensembles of genes interact to form an organism and maintain its vital systems.

Determining Gene Function

Once they identify a gene of interest, how do scientists determine its function? A gene's sequence can be compared with sequences in other species. If the function of a similar gene in another species is known, one might suspect that the gene product in question performs a comparable task. Data about the location and timing of gene expression may reinforce the suggested function. To obtain stronger evidence, one approach is to disable the gene and then observe the consequences in the cell or organism.

Editing Genes and Genomes

Molecular biologists have long sought techniques for altering, or editing, the genetic material of cells or organisms in a predictable way. In one such technique, called ***in vitro* mutagenesis**, specific mutations are introduced into a cloned gene, and the mutated gene is returned to a cell in such a way that it disables ("knocks out") the normal cellular copies of the same gene. If the introduced mutations alter or destroy the function of the gene product, the phenotype of the mutant cell may help reveal the function of the missing normal protein.

In Concept 17.5, you read about the CRISPR-Cas9 system, the powerful new technique for gene editing in living cells and organisms that is taking the field of genetic engineering by storm (see Figure 17.28). This system, worked out by Jennifer Doudna (**Figure 20.14**) and Emmanuelle Charpentier, is a highly effective way for researchers to knock out a given gene in order to study what that gene does. It has already been used in many organisms, including bacteria, fish, mice, insects, human cells, and various crop plants. Modifications of the technique allow researchers to repair a gene that has a mutation. This approach is used for gene therapy, which will be discussed later in the chapter.

In another application of the CRISPR-Cas9 system, scientists are attempting to address the global problem of insect-borne diseases by altering genes in insects so that,